

Ordinary Differential Equations: Boundary Value Problems

Numerical Analysis E3/I3, FMN050

Numerical Analysis

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Boundary value problems

Standard problems:

Two-point BVP (2p-BVP)

$$\begin{aligned}y'' &= f(x, y, y') \\ y(a) &= \alpha; \quad y(b) = \beta\end{aligned}$$

Find solution $y(x)$.

Eigenvalue problems:

Regular *Sturm–Liouville* problems

$$\begin{aligned}\frac{d}{dx} \left(p(x) \frac{dy}{dx} \right) + \lambda y &= 0 \\ y(a) &= 0; \quad y(b) = 0\end{aligned}$$

Find eigenfunctions $y(x)$ and eigenvalues λ .

Boundary value problems. . .

Two types of solution techniques:

- Direct discretization (“banded matrix method”)
- Shooting (use IVP solution techniques)

Direct discretization (finite difference methods):

Approximate the derivatives y' and y'' in

$$y'' = f(x, y, y')$$

$$y(a) = \alpha; \quad y(b) = \beta$$

by symmetric *finite difference* quotients.

Symmetry $\Rightarrow$ global error better than

$$y_n - y(x_n) = c_1 h^2 + c_2 h^4 + c_3 h^6 + \dots$$

Finite differences

Forward difference operator

$$\begin{aligned}\Delta y_i &= y_{i+1} - y_i \\ \Delta^k y_i &= \Delta(\Delta^{k-1} y_i)\end{aligned}$$

In particular, 2nd difference:

$$\begin{aligned}\Delta^2 y_i &= \Delta(y_{i+1} - y_i) \\ &= (y_{i+2} - y_{i+1}) - (y_{i+1} - y_i) \\ &= y_{i+2} - 2y_{i+1} + y_i\end{aligned}$$

Approximation of 2nd derivative:

$$\frac{d^2 y}{dx^2} \approx \frac{y_{i+1} - 2y_i + y_{i-1}}{\Delta x^2} = \frac{\Delta^2 y_{i-1}}{\Delta x^2}$$

Approximation of derivatives

First order approximations

Forward difference:

$$y'(x) = \frac{y(x + \Delta x) - y(x)}{\Delta x} + O(\Delta x)$$

Backward difference:

$$y'(x) = \frac{y(x) - y(x - \Delta x)}{\Delta x} + O(\Delta x)$$

Second order approximations

Symmetric difference quotients:

$$y'(x) = \frac{y(x + \Delta x) - y(x - \Delta x)}{2\Delta x} + O(\Delta x^2)$$

$$y''(x) = \frac{y(x + \Delta x) - 2y(x) + y(x - \Delta x)}{\Delta x^2} + O(\Delta x^2)$$

Derivatives and matrices

Forward difference:

$$y'(x) = \frac{y(x + \Delta x) - y(x)}{\Delta x} + O(\Delta x)$$

Introduce vectors $y = \{y(x_i)\}$ and $y' = \{y'(x_i)\}$:

$$\begin{pmatrix} y'_0 \\ y'_1 \\ \vdots \\ y'_N \end{pmatrix} \approx \frac{1}{\Delta x} \begin{pmatrix} -1 & 1 & & & \\ & -1 & 1 & & \\ & & \ddots & \ddots & \\ & & & -1 & 1 \end{pmatrix} \begin{pmatrix} y_0 \\ y_1 \\ \vdots \\ y_{N+1} \end{pmatrix}$$

Note: $(N + 1) \times (N + 2)$ matrix.

Nullspace spanned by $y = (1 \quad 1 \quad 1 \dots 1)^T$.

Compare nullspace of d/dx : $y = 1 \Rightarrow y' \equiv 0$.

Analogous result for backward difference.

Derivatives and matrices. . .

Central difference:

$$y'(x) \approx \frac{y(x + \Delta x) - y(x - \Delta x)}{2\Delta x}$$

$$\begin{pmatrix} y'_1 \\ y'_2 \\ \vdots \\ y'_N \end{pmatrix} \approx \frac{1}{2\Delta x} \begin{pmatrix} -1 & 0 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} y_0 \\ y_1 \\ \vdots \\ y_{N+1} \end{pmatrix}$$

Note: $N \times (N + 2)$ matrix with nullspace spanned by $\bar{y} = (1 \ 1 \ 1 \dots 1)^T$ and $\tilde{y} = (1 \ -1 \ 1 \ -1 \dots 1)^T$.

False nullspace: $\tilde{y} = (1 \ -1 \ 1 \ -1 \dots 1)^T$ *does not converge to a C^1 function!*

Compare difference equation $y_{n+1} - y_{n-1} = 0$ with characteristic equation

$$z^2 - 1 = 0 \quad \Rightarrow \quad z = \pm 1$$

and solutions $\bar{y}_n = 1$ and $\tilde{y}_n = (-1)^n$, respectively.

Derivatives and matrices. . .

Central difference:

$$y''(x) \approx \frac{y(x + \Delta x) - 2y(x) + y(x - \Delta x)}{\Delta x^2}$$

$$\begin{pmatrix} y_1'' \\ y_2'' \\ \vdots \\ y_N'' \end{pmatrix} \approx \frac{1}{\Delta x^2} \begin{pmatrix} 1 & -2 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & -2 & 1 \end{pmatrix} \begin{pmatrix} y_0 \\ y_1 \\ \vdots \\ y_{N+1} \end{pmatrix}$$

Note: $N \times (N + 2)$ matrix with nullspace spanned by $\bar{y} = (1 \ 1 \ 1 \dots 1)^T$ and $\hat{y} = (1 \ 2 \ 3 \ 4 \dots N)^T$.

Nullspace of d^2/dx^2 : $y = 1$ and $y = x$ have $y'' \equiv 0$.

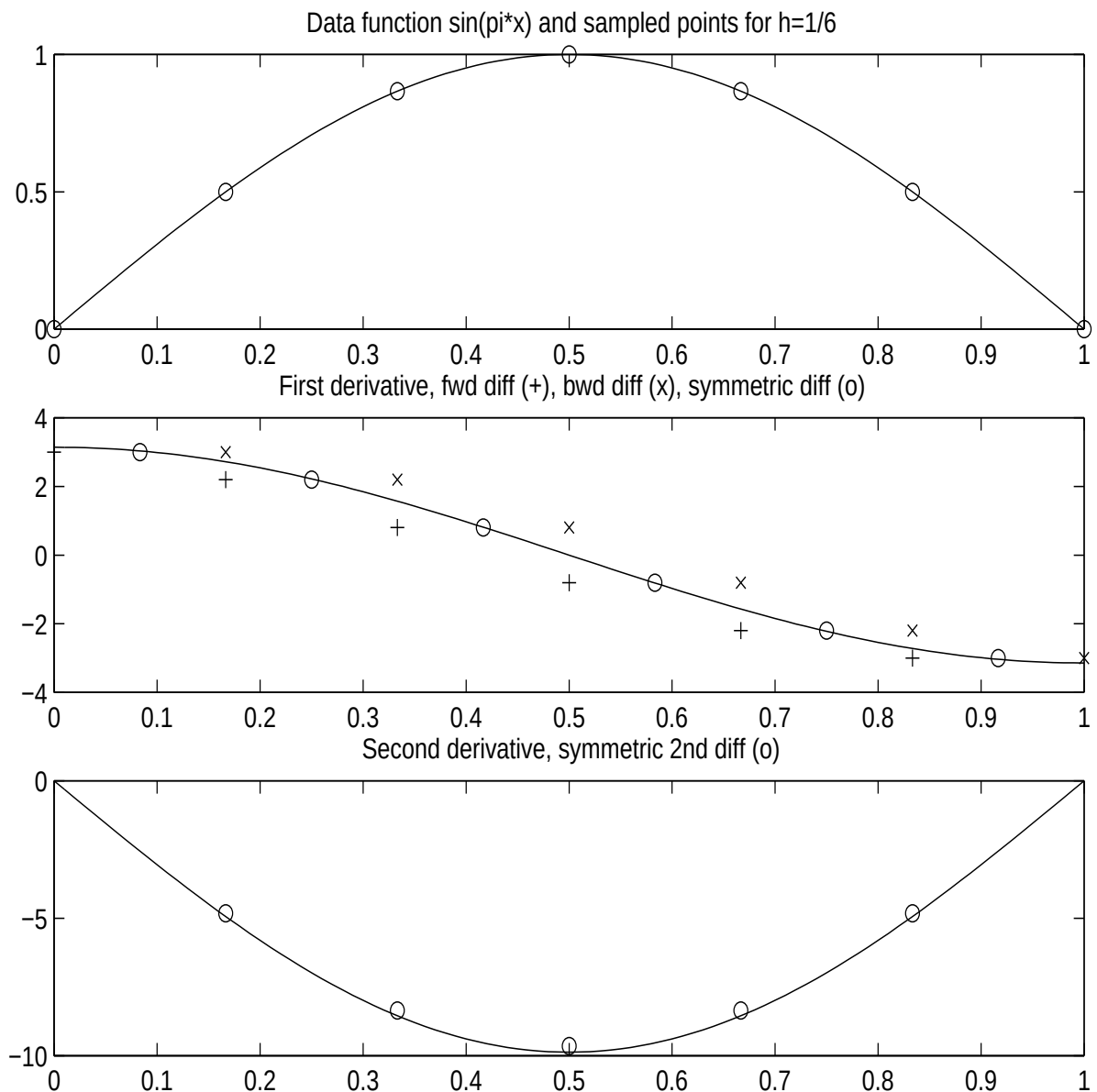
Compare difference equation $y_{n+1} - 2y_n + y_{n-1} = 0$ with characteristic equation

$$z^2 - 2z + 1 = 0 \quad \Rightarrow \quad z = 1, 1$$

and solutions $\bar{y}_n = 1$ and $\hat{y}_n = n$, respectively.

Numerical differentiation

First and second derivatives of $y = \sin \pi x$



Differential equations and discretization

Example:

$$y'' + y = 0$$
$$(D^2 + 1)y = 0$$

Homogeneous solutions: $y = \sin x$ and $y = \cos x$.

Discretization with central differences:

$$\left(\frac{1}{\Delta x^2} \text{tridiag} (1 \ -2 \ 1) + I \right) y = 0$$
$$(D_0^2 + I)y = 0$$

Homogeneous solution: find nullspace of $D_0^2 + I$!

Boundary conditions will determine “the” homogeneous solution!

Finite difference methods for 2p-BVP

Consider simplest problem:

$$\begin{aligned}y'' &= f(x, y) \\ y(0) &= \alpha; \quad y(1) = \beta\end{aligned}$$

Introduce equidistant grid with $\Delta x = 1/(N + 1)$.

Discretize:

$$\begin{aligned}\frac{y_{i+1} - 2y_i + y_{i-1}}{\Delta x^2} &= f(x_i, y_i) \\ y_0 &= \alpha; \quad y_{N+1} = \beta\end{aligned}$$

This is a (nonlinear) system of equations for the N unknowns $y_1, y_2, \dots, y_N$.

Solve $F(y) = 0$ using Newton's method.

Jacobian matrix

Equation system (note boundary values):

$$F_1(y) = \alpha - 2y_1 + y_2 - \Delta x^2 f(x_1, y_1)$$

$$F_i(y) = y_{i-1} - 2y_i + y_{i+1} - \Delta x^2 f(x_i, y_i)$$

$$F_N(y) = y_{N-1} - 2y_N + \beta - \Delta x^2 f(x_N, y_N)$$

The **Jacobian** matrix

$$F'(y) = \text{tridiag} \left(1, \quad -2 - \Delta x^2 \frac{\partial f}{\partial y_i}, \quad 1 \right)$$

is **tridiagonal**, and

- Superdiagonal elements 1
- Diagonal elements $-2 - \Delta x^2 \partial f / \partial y_i$
- Subdiagonal elements 1
- LU decomposition runs in $O(N)$ time
- Solution effort moderate even when N is large.

Local and global errors

Local error: Insert exact solution into discretization

$$\frac{y(x_{i-1}) - 2y(x_i) + y(x_{i+1}))}{\Delta x^2} = f(x_i, y(x_i)) - l(x_i)$$

Taylor series expansion, using $f(x_i, y(x_i)) = y''(x_i)$:

$$-l(x_i) = 2 \left(\frac{\Delta x^2}{4!} y^{(4)}(x_i) + \frac{\Delta x^4}{6!} y^{(6)}(x_i) + \dots \right)$$

Only **even** powers of Δx due to **symmetry**.

Global error: $e(x_i) = y_i - y(x_i)$.

Will show that

$$e(x_i) = c_1 \Delta x^2 + c_2 \Delta x^4 + \dots$$

Again only even powers due to symmetry.

Convergence

Assume f is a linear function, $\Rightarrow F$ is linear $\Rightarrow$

$$F(y) = 0 \quad \Leftrightarrow \quad T_h y = q$$

with $h = \Delta x$ and T_h tridiagonal. Then

Numerical solution: $T_h y = q$

Exact solution: $T_h y(x) = q - l(x)$

Solve the system (invert T_h) to get

Numerically: $y = T_h^{-1} q$

Exact: $y(x) = T_h^{-1} (q - l(x))$

Global error: $e(x) = T_h^{-1} l(x)$

Convergence: $\|e(x)\|_{\text{RMS}} \leq \|T_h^{-1}\|_2 \cdot \|l(x)\|_{\text{RMS}}$

where $\|l(x)\|_{\text{RMS}}^2 = \frac{1}{N} \sum_1^N l(x_i)^2$

Convergence. . .

Since $\|l\|_{\text{RMS}} = \gamma_1 h^2 + \gamma_2 h^4 \dots$

if $\|T_h^{-1}\|_2 \leq C$ **(independently of $h \rightarrow 0$) then**

$$\|e\|_{\text{RMS}} \leq C \cdot \|l\|_{\text{RMS}} = c_1 h^2 + c_2 h^4 + \dots$$

One can prove (difficult) that $\|T_h^{-1}\|_2 \lesssim 1/\pi^2$.

Conclusion:

Consistency: local error $l \rightarrow 0$ as $h \rightarrow 0$

Stability: $\|T_h^{-1}\|_2 \leq C$ as $h \rightarrow 0$

Convergence: global error $e \rightarrow 0$ as $h \rightarrow 0$

Lax' equivalence theorem:

$$\text{Consistency} + \text{Stability} \Rightarrow \text{Convergence}$$

(“Fundamental theorem of numerical analysis”)

Cowell's method

Standard symmetric discretization implies order $p = 2$.

Note:

$$-l(x_i) \approx \frac{h^2}{12} y^{(4)}(x_i) + O(h^4)$$

But $y'' = f \Rightarrow y^{(4)} = f''$, so approximate by

$$\frac{h^2}{12} y^{(4)}(x_i) \approx \frac{f_{i-1} - 2f_i + f_{i+1}}{12}$$

Add this correction to 2nd order discretization

$$\frac{y_{i-1} - 2y_i + y_{i+1}}{h^2} = f_i$$

to obtain **Cowell's method** (order $p = 4$):

$$\frac{y_{i-1} - 2y_i + y_{i+1}}{h^2} = \frac{1}{12} (f_{i-1} + 10f_i + f_{i+1})$$

Implicit difference correction. Jacobian still tridiagonal.

Sturm–Liouville problems

Diffusion problem

$$u_t = \operatorname{div} (p \operatorname{grad} u)$$

Separation of variables (one space dimension):

$$u(t, x) := y(x) \cdot v(t) \quad \Rightarrow$$

$$\frac{\dot{v}}{v} = \frac{\operatorname{div} (p(x) \operatorname{grad} y(x))}{y} =: -\lambda$$

Sturm–Liouville eigenvalue problem:

$$\frac{d}{dx} \left(p(x) \frac{dy}{dx} \right) + \lambda y = 0$$

$$y(a) = 0; \quad y(b) = 0.$$

Find eigenvalues λ and eigenfunctions $y(x)$.

Sturm–Liouville: discretization

$$\frac{d}{dx} \left(p(x) \frac{dy}{dx} \right) + \lambda y = 0$$
$$y(a) = 0; \quad y(b) = 0.$$

$$\frac{p_{i-1/2}y_{i-1} - (p_{i-1/2} + p_{i+1/2})y_i + p_{i+1/2}y_{i+1}}{h^2} + \lambda_h y_i = 0$$

$$y_0 = y_{N+1} = 0.$$

Symmetric tridiagonal matrix eigenvalue problem:

$$T_h y = -h^2 \lambda_h y$$

The discretization has N eigenvalues $\lambda_h = \lambda + O(h^2)$.

Sturm–Liouville: example

Simplest example:

$$\begin{aligned}y'' + \lambda y &= 0 \\ y(0) = y(1) &= 0\end{aligned}$$

General solution:

$$y(x) = A \sin \sqrt{\lambda} x + B \cos \sqrt{\lambda} x$$

Boundary values $\Rightarrow B = 0$ and

$$\begin{aligned}A \sin \sqrt{\lambda} &= 0 \\ A \neq 0 &\Rightarrow \sqrt{\lambda} = n\pi\end{aligned}$$

Eigenvalues: $\lambda = n^2\pi^2, \quad n = 1, 2, \dots$

Eigenfunctions: $y = \sin n\pi x$

Discrete Sturm–Liouville: example

Discretization of $y'' + \lambda y = 0$ with boundary values $y(0) = y(1) = 0$ yields

$$\frac{y_{i-1} - 2y_i + y_{i+1}}{h^2} + \lambda_h y_i = 0$$

$$y_0 = y_{N+1} = 0; \quad h = 1/(N + 1).$$

Tridiagonal $N \times N$ matrix formulation

$$\frac{1}{h^2} \begin{pmatrix} -2 & 1 & & \\ 1 & -2 & 1 & \\ & & \ddots & \\ & & 1 & -2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix} = -\lambda_h \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix}$$

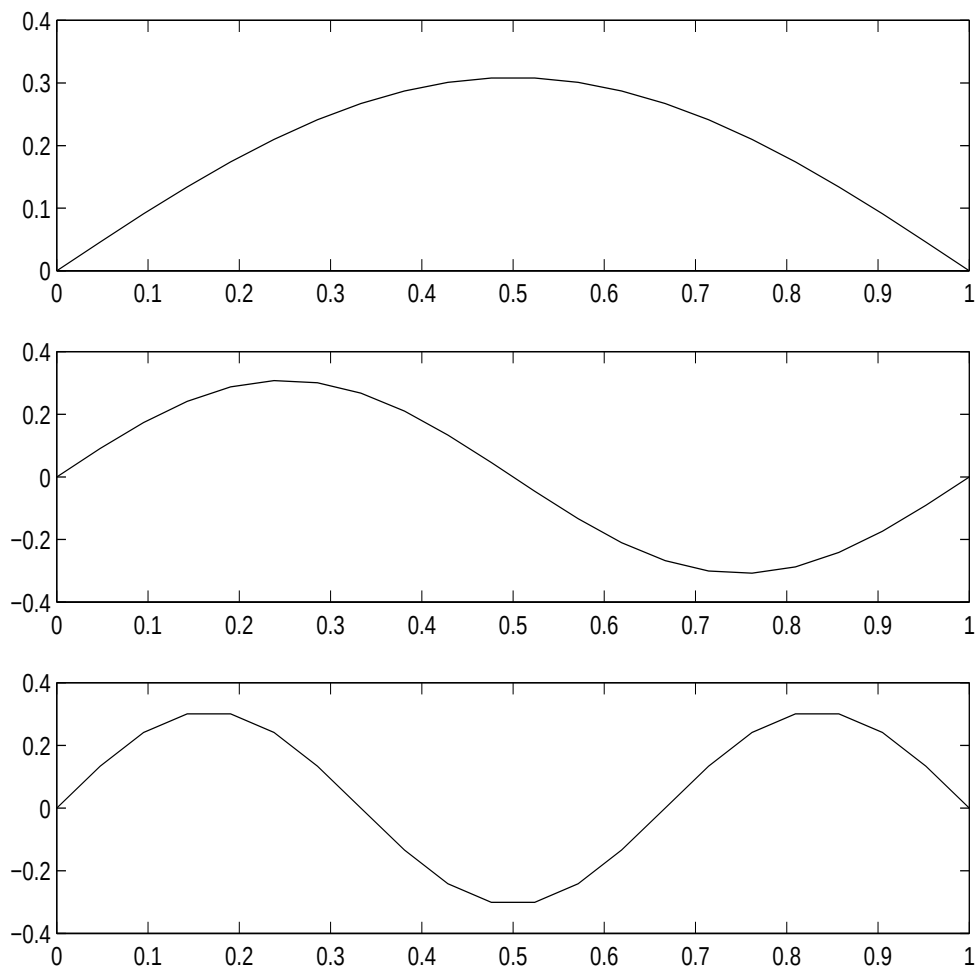
Algebraic eigenvalue problem

$$T_h y = -\lambda_h y$$

Smallest eigenvalue $\lambda_h = \pi^2 + O(h^2)$.

Discrete Sturm–Liouville: computation

Three first eigenvalues and their eigenvectors ($N = 20$)



Discrete e'values λ_h	9.8512	39.1848	87.3455
Exact e'values λ	9.8696	39.4784	88.8264
Relative errors	0.19%	0.74%	1.67%